FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

IEC 61499 Compliant Web-based IDE for Function Block Design

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List of Acronyms

CPPS Cyber-Physical Production Systems

FBD Function Block Diagram

ICS Industrial Control Systems

IDE Integrated Development Environment

PLC Programmable Logic Controller

RTE Runtime Environment

SOA Service-Oriented Architectures

UI User Interface

Chapter 1

Introduction

1.1 Context

Cyber-Physical Production Systems (CPPS) [3] are complex systems that integrate physical processes with digital technologies to optimise production processes. Several software engineering approaches and technologies are utilised in the development and operation of CPPS. One of such approaches is Function Block Diagram (FBD) [1], which is a graphical programming language used in industrial automation and control systems. FBD uses graphical function blocks to represent the logic and control of a system. The function blocks are connected in a network to form a control program. One of the benefits of FBD is that it provides a visual representation of the control logic, which can be easier to understand and debug than traditional textual programming languages. FBD is also modular, which means that function blocks can be reused in multiple programs, reducing development time and improving code maintainability. Also, FBD is often used in conjunction with Programmable Logic Controllers (PLCs) to control machinery and processes in manufacturing and other industrial applications [4, 5]. FBD programs can be created and edited using specialised software tools that allow for the creation and manipulation of function blocks [11].

In industrial automation, it is common to use the IEC 61499 standard, which defines an event-driven FBD way to design control software, especially for distributed systems [7].

While Runtime Environments (RTEs) enable the execution of function blocks compliant with the standard, Integrated Development Environments (IDEs) are used to design, create, and deploy function block pipelines to CPPS. These IDEs offer features and tools that facilitate the development, testing, and deployment of IEC 61499-compliant control systems. They provide graphical editors for creating function block diagrams, simulation capabilities, and additional functionalities to streamline the development process.

1.2 Motivation

The evolution of CPPS has increased the demand for flexible and intelligent control solutions capable of integrating physical processes with digital technologies. FBD and the IEC 61499 standard offer a structured, modular approach to building such distributed automation systems. To fully leverage these benefits, development tools must evolve alongside industrial needs, supporting more accessible, collaborative, and scalable workflows.

1.3 Problem

Existing IEC 61499 development environments are predominantly desktop-based and lack modern capabilities such as remote access, real-time collaboration, and seamless integration with cloud infrastructures. These limitations hinder efficient development, maintenance, and deployment of control applications within distributed CPPS. As a result, current tools fall short in supporting the flexibility, scalability required by certain modern industrial automation systems. Their complexity further complicates their usage for more basic needs.

1.4 Research Questions

This research seeks to investigate how modern web technologies can be leveraged to enhance the development, deployment, and operation of IEC 61499-compliant systems. This study is guided by the following main research question:

> How can web-based IDEs enhance the development, deployment, and collaboration processes for IEC 61499 applications in distributed CPPS, addressing the limitations of traditional desktop-based environments?

To further refine the scope, the following sub-questions are proposed:

- What architectural and technological requirements must be considered when designing a web-based IDEs for IEC 61499?
- In what ways do web-based IDEs improve collaboration, accessibility, and scalability compared to existing desktop tools?

Chapter 2

Literature Review

This chapter reviews prior work related to IEC 61499-based development, CPPS, and web-based IDEs. The goal is to identify how current tools support distributed automation and identify where gaps exist.

2.1 Methodology

A systematic literature review approach was adopted, the objective was to collect, screen, and analyze studies addressing IEC 61499 development tools, function blocks, CPPS, and web-based development environments, to then critically reflect upon their findings to help the writing of this dissertation and the planning of the final product.

2.1.1 Search and Selection

The search was conducted in digital libraries **ScienceDirect**, **IEEE Xplore**, **ACM Digital Library**, among others, as well as research paper aggregators like **Semantic Scholar**, **Scopus** and **Web of Science** using combinations of the following terms:

(`"IEC 61499"` OR `"FBD"`) AND (`"web-based"` OR `"IDE"`) AND (`"CPPS"`)

Following the search, a manual review process was conducted to refine and validate the results. Titles and abstracts were first screened to understand the relevance of the articles with respect to this dissertation. The publications were then examined in greater detail through full-text review, the reference lists of selected papers were manually inspected to identify relevant works that may not have been captured by the initial keyword search.

2.2 Related Work

Some recent studies explore web-based execution of FBD, Rohat and Popescu [6] as well as Lyu et al. [2] propose the creation of a web platform for IEC 61499 development. While their works are important case studies that will be taken into account and thoroughly studied, it should be of note that this paper has different objectives and aims to achieve a different final product.

Chapter 3

State of the Art

This chapter provides a comprehensive analysis of the current landscape of development environments for CPPS and the IEC 61499 standard. It categorizes existing solutions into desktop-based and web-based development tools, while analyzing a possible gap in the ecosystem.

3.1 Development Ecosystem

Currently, the development ecosystem is dominated by desktop-based IDEs. These tools are typically monolithic applications installed locally on engineering workstations.

3.1.1 Open-Source

The most prominent open-source implementation is the **Eclipse 4diac** framework. It consists of the **4diac IDE** (based on the **Eclipse** framework) and the **4diac FORTE** (a run-time environment), although it can also use different RTEs like **DINASORE**.

This IDE is very feature rich at the cost of significant system resources and increased complexity which make it more difficult to use. It relies on the heavyweight **Eclipse Rich Client Platform**.

3.1.2 Commercial

The commercial side is dominated by **EcoStruxure Automation Expert**, **ISaGRAF Workbench** and other monolithic desktop based environments, usually with a built-in proprietary RTE. This makes them often very complete production ready tools, but they are not very flexible and not the best fit for many needs.

3.2 Evolution of Web-Based Engineering Tools

The software engineering domain has seen a massive shift from desktop IDEs to cloud-native environments. This trend is now influencing the domain of Industrial Control Systems (ICS), and while recent studies have proposed a Service-Oriented Architectures (SOA) to integrate IEC 61499 with cloud layers [8], the engineering tools themselves remain largely desktop-bound.

3.2.1 Web-Based IDEs in Software Development

In general software development, tools like **Visual Studio Code** have started offering web-based counterparts to their desktop applications [9] in search of a more flexible user experience. In fact, newer development environments like **Zed** make it a key goal of their products to have web platform support [10].

3.3 Gap Analysis

Current IEC 61499 tools face significant limitations that hinder modern CPPS development:

- **Platform Dependency:** Existing solutions are monolithic desktop applications, lacking the accessibility and portability of web-based environments.
- **Resource Intensity:** Current options are heavyweight and complex, creating a high barrier to entry compared modern development tools of other purposes.
- **Web Trends:** Despite industry trends, there is currently no mature, lightweight web interface for FBD, leaving ICS engineering behind the curve of cloud-native software development.

Chapter 4

Proposed Solution

4.1 Approach

4.2 Methods

4.2.1 Introduction

4.3 Workplan

Figure 4.1 outlines the planned work for this dissertation. The schedule is organized into three main phases: Experimentation, Code Development, and Deliveries. During the Experimentation phase, there will be a focus on studying the standard and running experiments to establish a solid technical foundation. The Code Development phase overlaps partially with Experimentation, ensuring that the software evolves in parallel with the experimental insights. Finally, the Deliveries phase spans the entire project duration and includes writing the dissertation and preparing the final presentation, ensuring that all outputs are documented and communicated effectively.

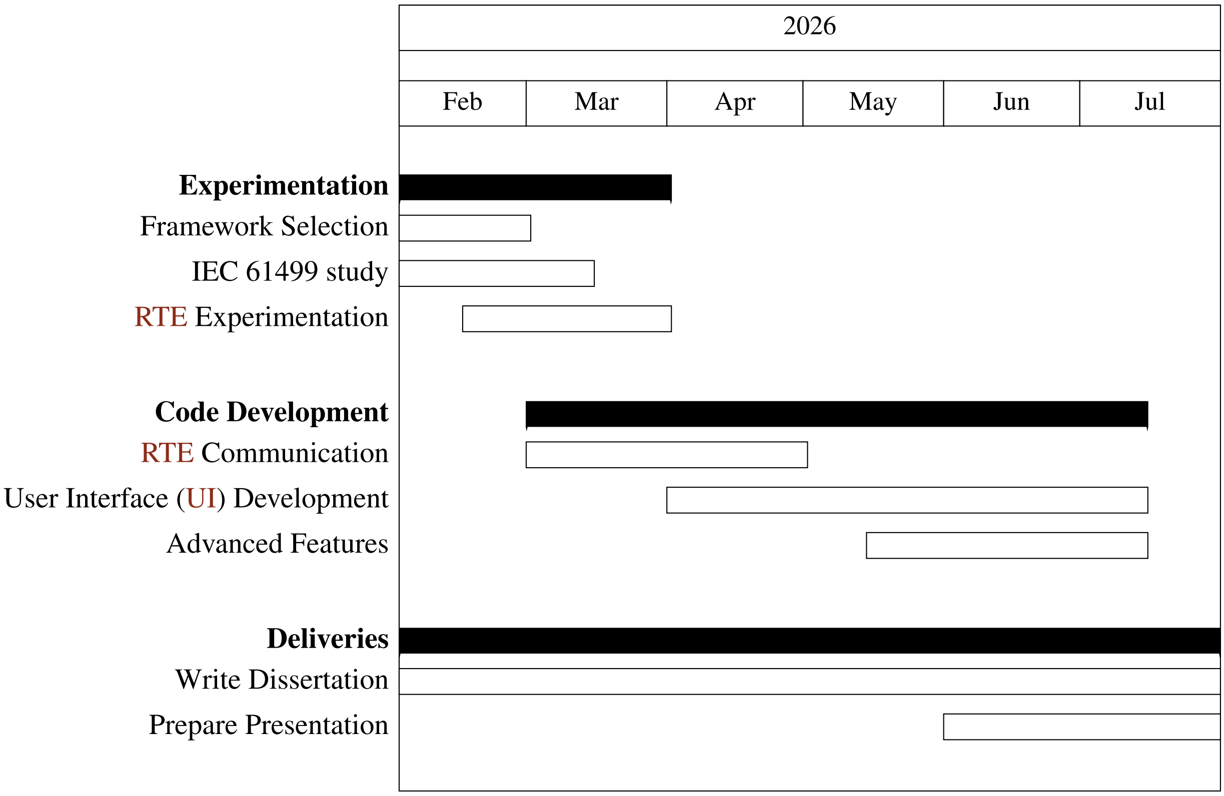

Figure 4.1: Gantt chart timeline for the proposed work

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